

Appointments at Mayo Clinic

Mayo Clinic offers appointments in Arizona, Florida and Minnesota and at Mayo Clinic Health System locations.

[Request Appointment](#)

First aid

Print

Cardiopulmonary resuscitation (CPR): First aid

[Basics](#) [In-Depth](#) [Multimedia](#) [Resources](#)

Products and services

The Mayo Clinic Diet

What is your weight-loss goal?

5 - 10 lbs »

11 - 25 lbs »

25+ lbs »

Learn the steps to perform this lifesaving technique on adults and children.

[By Mayo Clinic Staff](#)

Cardiopulmonary resuscitation (CPR) is a lifesaving technique that's useful in many emergencies in which someone's breathing or heartbeat has stopped. For example, when someone has a heart attack or nearly drowns. The American Heart Association recommends starting CPR with hard and fast chest compressions. This hands-only CPR recommendation applies to both untrained bystanders and first responders.

Advertisement

Mayo Clinic Press

Check out these best-sellers and special offers on books and newsletters from [Mayo Clinic Press](#).

[Mayo Clinic on Incontinence](#)

[The Essential Diabetes Book](#)

[Mayo Clinic on Hearing and Balance](#)

[FREE Mayo Clinic Diet Assessment](#)

[Mayo Clinic Health Letter - FREE book](#)

Advertisement



Mayo Clinic does not endorse companies or products. Advertising revenue supports our not-for-profit mission.

Advertising & Sponsorship

[Policy](#) | [Opportunities](#) | [Ad Choices](#)

If you're afraid to do CPR or unsure how to perform CPR correctly, know that it's always better to try than to do nothing at all. The difference between doing something and doing nothing could be someone's life.

Here's advice from the American Heart Association:

- **Untrained.** If you're not trained in CPR or worried about giving rescue breaths, then provide hands-only CPR. That means uninterrupted chest compressions of 100 to 120 a minute until paramedics arrive (described in more detail below). You don't need to try rescue breathing.
- **Trained and ready to go.** If you're well-trained and confident in your ability, check to see if there is a pulse and breathing. If there is no pulse or breathing within 10 seconds, begin chest compressions. Start CPR with 30 chest compressions before giving two rescue breaths.
- **Trained but rusty.** If you've previously received CPR training but you're not confident in your abilities, then just do chest compressions at a rate of 100 to 120 a minute. Details are described below.

The above advice applies to situations in which adults, children and infants need CPR, but not newborns. Newborns are babies up to 4 weeks old.

CPR can keep oxygen-rich blood flowing to the brain and other organs until emergency medical treatment can restore a typical heart rhythm. When the heart stops, the body no longer gets oxygen-rich blood. The lack of oxygen-rich blood can cause brain damage in only a few minutes.

If you are untrained and have immediate access to a phone, call 911 or your local emergency number before beginning CPR. The dispatcher can tell you how to do the proper procedures until help arrives. To learn CPR properly, take an accredited first-aid training course, including CPR and how to use an automated external defibrillator (AED).

Before you begin

Before starting CPR, check:

- Is the environment safe for the person?
- Is the person conscious or unconscious?
- If the person appears unconscious, tap or shake their shoulder and ask loudly, "Are you OK?"
- If the person doesn't respond and you're with another person who can help, have one person call 911 or the local emergency number and get the AED, if one is available. Have the other person begin CPR.
- If you are alone and have immediate access to a telephone, call 911 or your local emergency number before beginning CPR. Get the AED if one is available.
- As soon as an AED is available, deliver one shock if instructed by the device, then begin CPR.

Remember to spell C-A-B

The American Heart Association uses the letters C-A-B to help people remember the order to perform the steps of CPR.

- **C:** compressions
- **A:** airway
- **B:** breathing

Compressions: Restore blood flow



Compressions means you use your hands to push down hard and fast in a specific way on the person's chest. Compressions are the most important step in CPR. Follow these steps for performing CPR compressions:



Rescue breathing

1. Put the person on their back on a firm surface.
2. Kneel next to the person's neck and shoulders.
3. Place the lower palm of your hand over the center of the person's chest, between the nipples.
4. Place your other hand on top of the first hand. Keep your elbows straight. Place your shoulders directly above your hands.
5. Push straight down on the chest at least 2 inches (5 centimeters) but no more than 2.4 inches (6 centimeters). Use your entire body weight, not just your arms, when doing compressions.
6. Push hard at a rate of 100 to 120 compressions a minute. The American Heart Association suggests performing compressions to the beat of the song "Stayin' Alive." Allow the chest to spring back after each push.
7. If you haven't been trained in CPR, continue chest compressions until there are signs of movement or until emergency medical personnel take over. If you have been trained in CPR, go on to opening the airway and rescue breathing.

Airway: Open the airway

If you're trained in CPR and you've performed 30 chest compressions, open the person's airway using the head-tilt, chin-lift maneuver. Put your palm on the person's forehead and gently tilt the head back. Then with the other hand, gently lift the chin forward to open the airway.

Breathing: Breathe for the person

Rescue breathing can be mouth-to-mouth breathing or mouth-to-nose breathing if the mouth is seriously injured or can't be opened. Current recommendations suggest performing rescue breathing using a bag-mask device with a high-efficiency particulate air (HEPA) filter.

1. After opening the airway (using the head-tilt, chin-lift maneuver), pinch the nostrils shut for mouth-to-mouth breathing and cover the person's mouth with yours, making a seal.
2. Prepare to give two rescue breaths. Give the first rescue breath — lasting one second — and watch to see if the chest rises.
3. If the chest rises, give a second breath.
4. If the chest doesn't rise, repeat the head-tilt, chin-lift maneuver and then give a second breath. Thirty chest compressions followed by two rescue breaths is considered one cycle. Be careful not to provide too many breaths or to breathe with too much force.
5. Continue chest compressions to restore blood flow.

6. As soon as an automated external defibrillator (AED) is available, apply it and follow the prompts. Give one shock, then continue chest compressions for two more minutes before giving a second shock. If you're not trained to use an AED, a 911 operator or another emergency medical operator may be able to give you instructions. If an AED isn't available, go to step 5 below.
7. Continue CPR until there are signs of movement or emergency medical personnel take over.

To perform CPR on a child

The procedure for giving CPR to a child age 1 through puberty is essentially the same as that for an adult — follow the C-A-B steps. The American Heart Association says you should not delay CPR and offers this advice on how to perform CPR on a child:

Compressions: Restore blood flow

If you are alone and didn't see the child collapse, start chest compressions for about two minutes. Then quickly call 911 or your local emergency number and get the AED if one is available.

If you're alone and you did see the child collapse, call 911 or your local emergency number first. Then get the AED, if available, and start CPR. If another person is with you, have that person call for help and get the AED while you start CPR.

1. Place the child on their back on a firm surface.
2. Kneel next to the child's neck and shoulders.
3. Place two hands — or only one hand if the child is very small — on the lower half of the child's breastbone.
4. Using the heel of one or both hands, press straight down on the chest about 2 inches (approximately 5 centimeters) but not greater than 2.4 inches (approximately 6 centimeters). Push hard and fast — 100 to 120 compressions a minute.
5. If you haven't been trained in CPR, continue chest compressions until the child moves or until emergency medical personnel take over. If you have been trained in CPR, open the airway and start rescue breathing.

Airway: Open the airway

If you're trained in CPR and you've performed 30 chest compressions, open the child's airway using the head-tilt, chin-lift maneuver.

- Place your palm on the child's forehead and gently tilt their head back.
- With the other hand, gently lift the chin forward to open the airway.

Breathing: Breathe for the child

Follow these steps for mouth-to-mouth breathing for a child.

1. After using the head-tilt, chin-lift maneuver to open the airway, pinch the child's nostrils shut. Cover the child's mouth with yours, making a seal.
2. Breathe into the child's mouth for one second. Watch to see if the chest rises. If it rises, give a second breath. If the chest doesn't rise, repeat the head-tilt, chin-lift maneuver first. Then give the second breath. Be careful not to provide too many breaths or to breathe with too much force.

3. After the two breaths, immediately begin the next cycle of compressions and breaths. Note: If there are two people available to do CPR on the child, change rescuers every two minutes — or sooner if the rescuer is fatigued — and give one to two breaths every 15 compressions.
4. As soon as an AED is available, apply it and follow the prompts. As soon as an AED is available, apply it and follow the prompts. Use pediatric pads for children older than 4 weeks old and up to age 8. If pediatric pads aren't available, use adult pads. Give one shock, then restart CPR — starting with chest compressions — for two more minutes before giving a second shock. If you're not trained to use an AED, a 911 operator or another emergency medical operator may be able to give you directions.

Continue until the child moves or help arrives.

To perform CPR on a baby 4 weeks old or older

Cardiac arrest in babies is usually due to a lack of oxygen, such as from choking. If you know that the baby has an airway blockage, perform first aid for choking. If you don't know why the baby isn't breathing, perform CPR.

First, evaluate the situation. Touch the baby and watch for a response, such as movement. Don't shake the baby.

If there's no response, call 911 or your local emergency number, then immediately start CPR.

Follow the compressions, airway and breathing method for a baby under age 1. Do not follow this procedure for newborns, which include babies up to 4 weeks old.

If you saw the baby collapse, get the AED, if one is available, before starting CPR. If another person is available, have that person call for help immediately and get the AED while you stay with the baby and perform CPR.

Compressions: Restore blood flow

1. Place the baby on their back on a firm, flat surface, such as a table or floor.
2. Imagine a horizontal line drawn between the baby's nipples. Place two fingers of one hand just below this line, in the center of the chest.
3. Gently compress the chest about 1.5 inches, which is about 4 centimeters.
4. Count aloud as you push in a fairly rapid rhythm. You should push at a rate of 100 to 120 compressions a minute, just as you would when giving an adult CPR.

Airway: Open the airway

After 30 compressions, gently tip the head back by lifting the chin with one hand and pushing down on the forehead with the other hand.

Breathing: Breathe for the baby

1. Cover the baby's mouth and nose with your mouth.
2. Prepare to give two rescue breaths. Use the strength of your cheeks to deliver gentle puffs of air, instead of deep breaths from your lungs. Gently puff a breath into the baby's mouth

one time, taking one second for the breath. Watch to see if the baby's chest rises. If it does, give a second rescue breath. If the chest does not rise, repeat the head-tilt, chin-lift maneuver and then give the second breath.

3. If the baby's chest still doesn't rise, continue chest compressions.
4. Give two breaths after every 30 chest compressions. If two people are doing CPR, give one to two breaths after every 15 chest compressions.
5. Continue CPR until you see signs of life or until medical personnel arrive.

From Mayo Clinic to your inbox

Sign up for free and stay up to date on research advancements, health tips, current health topics, and expertise on managing health. [Click here for an email preview.](#)

Enter your email

[Learn more about Mayo Clinic's use of data.](#)



Subscribe!



Share



Tweet

July 27, 2023

Advertisement



Mayo Clinic does not endorse companies or products. Advertising revenue supports our not-for-profit mission.

Advertising & Sponsorship

[Policy](#) | [Opportunities](#) | [Ad Choices](#)

[Show references](#) ▾

See also

[Automated external defibrillators: Do you need an AED?](#)

[Marathon CPR Saves Life](#)

One gift, 3X the impact

Join our Year-End Challenge and triple your impact to help shape the future of healthcare!

Feedback

Give Now



Find a doctor



Explore careers



Sign up for free e-newsletters



About Mayo Clinic >

[About this Site](#)

[Contact Us](#)

[Locations](#)

[Health Information Policy](#)

[Media Requests](#)

[News Network](#)

Medical Professionals >

[AskMayoExpert](#)

[Clinical Trials](#)

[Mayo Clinic Alumni Association](#)

[Refer a Patient](#)

Businesses >

[Executive Health Program](#)

[International Business Collaborations](#)

[Facilities & Real Estate](#)

[Supplier Information](#)

Students >

[Admissions Requirements](#)

[Degree Programs](#)

[Student & Faculty Portal](#)

Researchers >

[Research Faculty](#)

[Laboratories](#)

International Patients >

[Appointments](#)

[Financial Services](#)

[International Locations & Offices](#)

Charitable Care & Financial Assistance >

[Community Health Needs Assessment](#)

[Financial Assistance Documents – Arizona](#)

[Financial Assistance Documents – Florida](#)

[Financial Assistance Documents – Minnesota](#)

Follow Mayo Clinic



Get the Mayo Clinic app



[Terms and Conditions](#)

[Privacy Policy](#)

[Notice of Privacy Practices](#)

[Notice of Nondiscrimination](#)

[Accessibility Statement](#)
[Advertising and Sponsorship Policy](#)
[Site Map](#)

Language: [English](#) ▼

© 1998-2023 Mayo Foundation for Medical Education and Research (MFMER). All rights reserved.